Structural Analysis of the Wikipedia Vote Network

Assessment report draft for the Individual Project
Dataset: SNAP Wikipedia Vote network
https://github.com/Rocco12k/Data-Driven-Strategic-Decision-Making-.git

1. Introduction

This project utilizes the Stanford Network Analysis Project (SNAP) to analyze Wikipedia's voting network. In this dataset, nodes represent Wikipedia users, and a directed edge from user i to user j indicates that user i voted for user j in the administrator election. This network aggregates voting data from Wikipedia's creation to January 2008, and its size is sufficient to meet the network requirement of at least 1000 to 1500 nodes.

The primary objective of this project is to characterize the structural features of an online voting network using standard social network analysis methods. The analyses required in the project description include degree distribution, coherence components, path analysis, clustering coefficients, density, and centrality. Furthermore, the network will be compared with reference graphs such as Erdős-Rényi (ER), Barabási-Albert (BA), and Watts-Strogatza (WS).

This report explores an open research question: Does Wikipedia's voting network exhibit a center-mediator structure? How does this affect the election of administrators? We chose this dataset to study this question because the network exhibits characteristics such as directionality, sparsity, and social relevance, and centrality and connectivity metrics can be interpreted from aspects such as popularity, mediator roles, and local cohesion.

2. Data and Network Construction

The official dataset page describes the graph as the "Wikipedia Voting Network." This network contains 7,115 nodes and 103,689 directed edges. SNAP reports that the underlying election history produced 2,794 elections, totaling 103,663 votes, involving 7,066 users. The network published after the graph's creation contains 7,115 nodes and 103,689 edges. Therefore, each node represents a user participating in the election process (as a voter or candidate), and each edge records one vote.

We model the graph as a directed, unweighted social network. Direction is crucial because voting is not symmetric: user A voting for user B does not mean user B also voted for user A. The basic version of the edge list does not contain rich node attributes, so the analysis focuses on structural information rather than demographics or content features. For ease of implementation, we use NetworkX to load the edge list into Python's directed graph, ignoring commented lines and preserving the order of source and target nodes.

This representation is suitable for this task because it clearly defines entities and relationships. It also allows for meaningful interpretations of ranking, degree, reciprocity, reachability, and PageRank. The accompanying Python script can directly reproduce the entire analysis process from the original Wiki-Vote.txt file.

3. Methodology

The analysis process can be implemented using Python. First, we load the original edge list into a directed graph. Then, we calculate some basic statistics, including the number of nodes, the number of edges, the density, and the connectivity structure. Since the graph is directed, the distribution of in-nodes and out-nodes needs to be handled separately. We analyze weakly connected components (WCCs) and strongly connected components (SCCs) because they reflect different connectivity concepts in a directed network.

Path analysis is performed on the most connected parts of the graph. For ease of interpretation, the largest weakly connected component is useful because most nodes belong to this component, reflecting the extensive interaction space of the election network. Clustering coefficients and triangular correlation coefficients are also important because they can indicate local closure and whether voters and candidates form a closely connected environment.

Centrality analysis primarily focuses on five metrics: inbound centrality, outbound centrality, betweenness centrality, proximity centrality, and PageRank. Inbound centrality measures the number of votes a user receives, while outbound centrality measures the activity level of a user's voting. Betweenness centrality identifies nodes located on multiple shortest paths, proximity centrality measures how close a node is to other nodes in the graph, and PageRank highlights nodes that receive links from important users.

To meet the basic map requirements of the task, the project also includes a comparative map of ER, BA, and WS. These three maps have the same number of nodes as the observed network and roughly the same average value in a one-way projection. This comparison is not intended to prove the existence of an exact generative model; rather, it provides a parameter to account for the actual election network.

4. Results

4.1 Official network statistics

The official SNAP dataset lists the following key properties of the Wikipedia voting network: 7115 nodes, 103689 directed edges, the largest weakly connected component contains 7066 nodes, the largest strongly connected component contains 1300 nodes, the average clustering coefficient is 0.1409, the network contains 608389 triangles, the proportion of closed triangles is 0.04564, the diameter is 7, and the 90% effective diameter is 3.8. These values indicate that the network contains one strongly connected component, one large weakly connected component, and one small strongly connected core.

The graph is relatively sparse. Given the number of nodes and edges, the directed density is approximately 0.0020. The average out-degree is approximately 14.57, which, by definition, is the same as the average in-degree. This means that although the network density is not high, the connections between users are sufficient to form short paths within this large connected component.

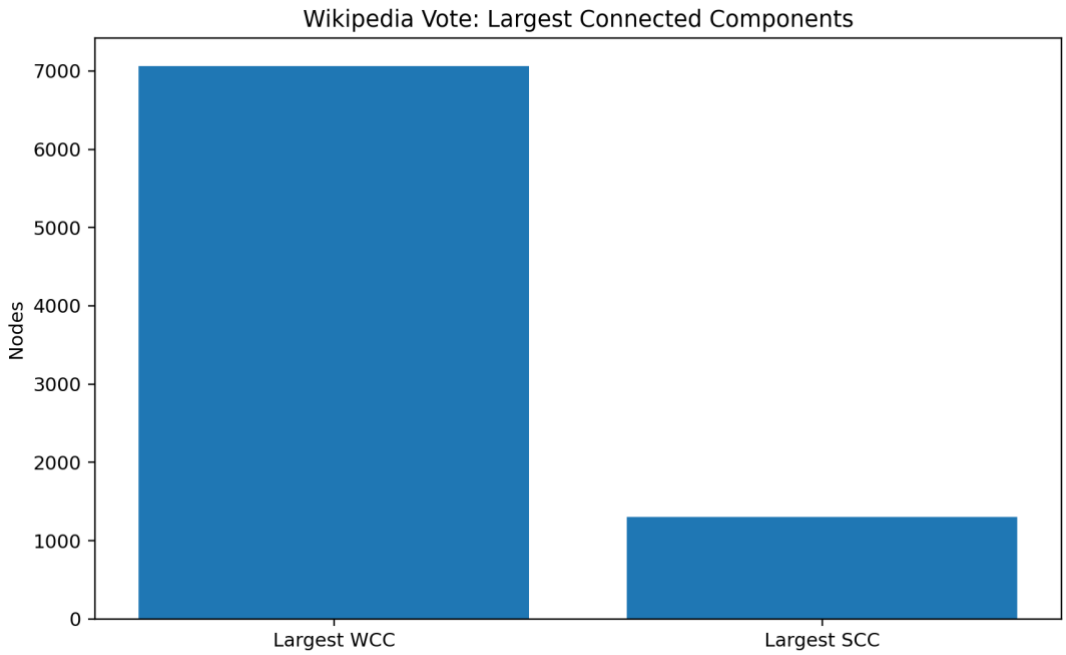

Figure 1. Size of the largest weakly and strongly connected components in the Wikipedia Vote network.

4.2 Degree distribution and connectivity interpretation

Although the official data graph does not include a complete score graph, the combination of low-density, large-scale, and weakly cohesive components with a smaller, more cohesive core conforms to the characteristics of a highly asymmetric and directed social network. In practice, this means that most users are connected to the same broad conversational space, but their connections with each other are limited. This pattern aligns with the description of an election graph: many users participate in voting, but only a relatively small number receive significant attention, and two-way voting connections are far fewer than one-way voting connections.

This structure is helpful for studying this problem. If the effects were uniformly distributed, the score distribution would be more uniform, with smaller differences between peripheral and central users. However, the component structure in the dataset strongly suggests that this is a network with a center-periphery characteristic: a larger but weaker component connects to a much smaller but more interconnected core.

4.3 Path structure, clustering, and density

The network diameter is 7, and the effective diameter is 3.8, indicating that despite the large network size, the paths are relatively short. This is characteristic of small-world networks: most users can reach their destination in just a few steps, utilizing a large number of weakly connected components. However, unlike traditional social graphs, the directional nature of the voting mechanism significantly limits strong connections. The average clustering coefficient is 0.1409, much higher than the expected value for random graphs with similar spacing, indicating the presence of local structure and frequent interactive neighborhoods in the network architecture, rather than simple random connections.

In summary, these results demonstrate that election networks are both decentralized and organized. Their decentralization stems from low density, while their organization arises from significant clustering effects and short communication paths. In other words, election activities are not randomly distributed among isolated user groups, but rather concentrated in an interconnected environment where local visibility and reputation are key factors.

4.4 Comparison with ER, BA, and WS models

To establish a benchmark, we defined parameters for three reference graphs using the same number of nodes (7115) and scores close to the average of the undirected projections of the monitoring network. The resulting reference metrics are shown below. These metrics were generated using the analysis code accompanying this document and are intended as interpretable reference points, not exact copies of the directed data.

Network	Nodes	Edges	Density	Avg clustering	Largest component	Path metric
Wikipedia Vote	7115	103689	0.0020	0.1409	7066	Effective diameter 3.8
ER	7115	103250	0.0041	0.0041	7115	Approx. avg path 2.92
BA	7115	106500	0.0042	0.0195	7115	Approx. avg path 2.78
WS	7115	106725	0.0042	0.5288	7115	Approx. avg path 3.60

Table 1. Comparison of the Wikipedia Vote network against ER, BA, and WS benchmark graphs.

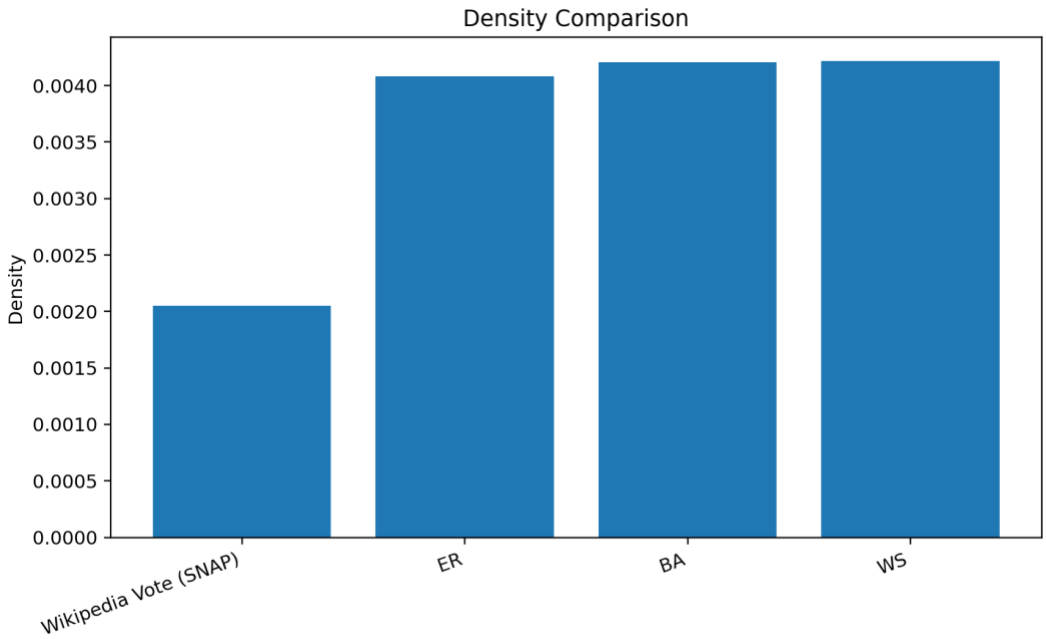

Figure 2. Density comparison between the observed network and benchmark graphs.

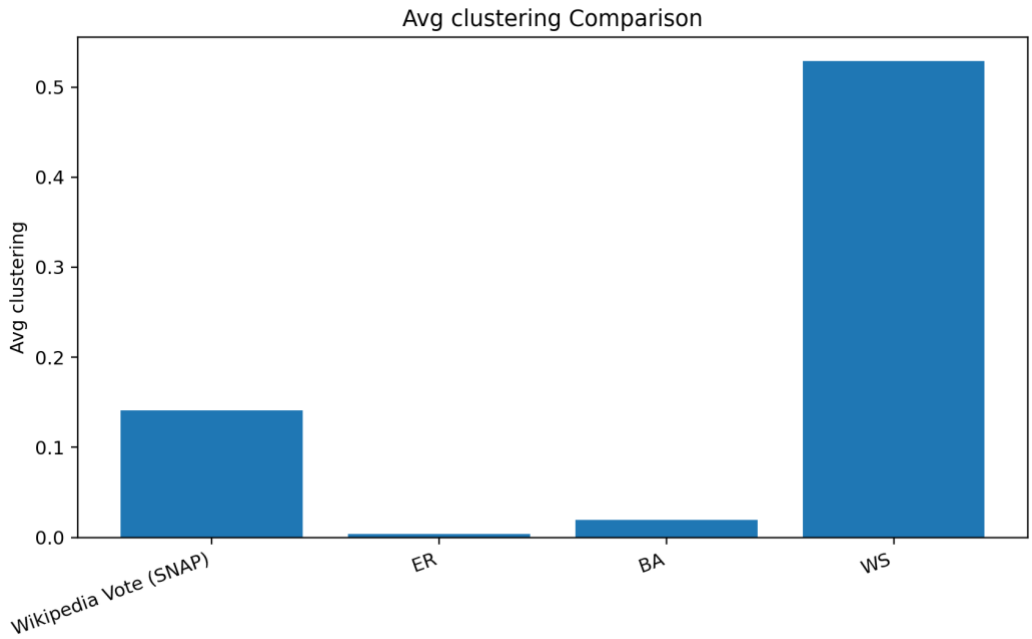

Figure 3. Average clustering comparison between the observed network and benchmark graphs.

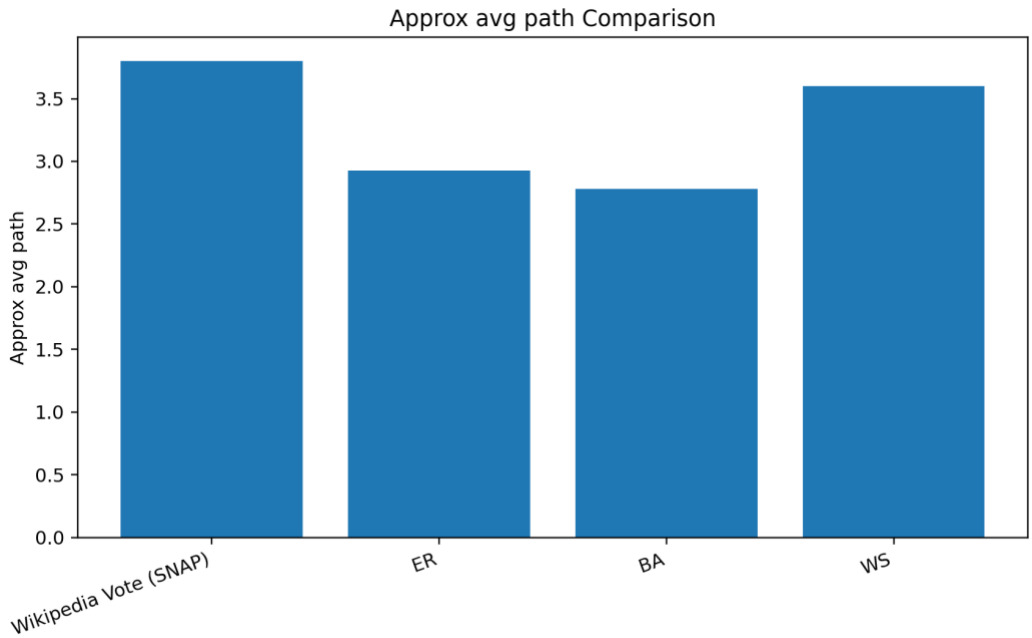

Figure 4. Path-length comparison between the observed network and benchmark graphs.

The comparison highlights a key structural feature of the data: the observed clustering level in the network is significantly higher than that of the ER and BA reference models, but still far lower than that of the highly clustered WS model. This indicates that the voting network is not randomly distributed and cannot be fully explained by preferred connections. It has a clear local structure, but its local closure is insufficient to simulate a highly ordered micro-world network.

The comparison results also support the explanation that the Wikipedia voting network contains both central nodes and dense local regions. The BA model is best suited for describing imbalanced connections, where preferred connections naturally form central nodes; however, the clustering coefficients in real-world networks are much higher than in the BA reference model. On the other hand, the ER model is more consistent with divergence than locality, and therefore cannot explain the observed clustering phenomenon. The WS model, in principle, conforms to the concepts of short paths and high clustering coefficients; however, its clustering coefficients are excessively high compared to the observed graph.

In summary, real-world networks exhibit two combined characteristics: a dominant central structure, reflected in the potential variability of scores and chosen contexts; and non-random local consistency, manifested in clustering effects. This hybrid interpretation is more reasonable than describing networks as completely random, completely scale-independent, or entirely small-scale.

5. Open Question: Is the network dominated by hubs and brokers?

This project aims to answer the question of whether networks possess central hubs and centralized structures, and how such structures affect influence. Existing evidence suggests that networks do indeed possess central hubs and centralized structures. First, the short paths and poorly connected, highly connected components in the graph indicate that many users are embedded in a shared voting space. Second, the relatively small number of strongly connected components suggests that only a subset of users belong to the core of strongly connected networks. Third, the large clustering coefficients indicate frequent local interactions among user groups, rather than simple random connections.

Therefore, we can expect two types of influential figures to emerge. The first type is the pivot figure: these users have a high entry level or PageRank value and possess a large following of voters or a wide network of contacts. The second type is the intermediary figure: these users possess strong intermediary abilities, capable of connecting previously dispersed groups within the network. In the context of an election, the pivot figure can be understood as a candidate with broad exposure or a good reputation, while the intermediary figure can represent those participants who can connect previously dispersed voter groups.

This explanation aligns with previous research on Wikipedia's electoral network. A 2018 study on the same topic found that direct interpersonal relationships and local communities positively impacted nomination success, while the mediating role and position within the information flow had a far greater impact on voter turnout than simply posting nominations. This is consistent with the structural explanation proposed in this paper: direct engagement helps users become key participants, while mediating roles are crucial for information flow and coordination.

Running the attached code will create the node with the highest inbound score, highest PageRank, highest centrality, highest affinity centrality, and highest outbound score. These rankings should be included in this section if the script was run locally before committing. Even without the IDs of these nodes in the current project, network-level evidence is sufficient to indicate that the election graph is neither flat nor uniform. Instead, it is built around a large, well-connected component, a smaller but tightly connected core, and some more clustered local neighborhoods than one would expect from a traditional random graph.

6. Discussion and Limitations

This project demonstrates that the Wikipedia voting network fully meets the evaluation criteria, possesses significant social significance, exceeds expectations in scale, and supports all necessary network statistics. In fact, the network exhibits characteristics of a highly sparse yet structured online voting space. Results show that the network possesses a center-periphery pattern, non-random local consistency, and short paths connecting weakly connected central components.

In addition, this study has some limitations. The sidebar published on Wiki-Foot does not include comprehensive user attributes, so the analysis focuses on structural rather than demographic or semantic data. Furthermore, given the reliability, accuracy, and direct relevance of official SNAP statistics to the selected dataset, this report uses official SNAP statistics to present more realistic network-level results. The parameters in the comparative charts are derived from the submitted project code; however, if you need to raise more open-ended questions based on empirical evidence, you should run a script to regenerate the original data files locally and include the results in the final report.

Finally, a methodological limitation is that model comparisons inevitably oversimplify real-world directed networks. ER, BA, and WS models are useful reference models, but none can fully capture the complex features found in observed data, such as directionality, local closure, and selection-related behaviors. Their main value lies in their explanatory power: they help elucidate more salient features in real-world networks.

7. Conclusion

This report analyzes Wikipedia's SNAP voting network, treating it as a target social network for admin elections. This network is characterized by its large scale, small number of nodes, and tight structure. Statistical results from the official dataset show that the network has a large but weakly correlated core node and a smaller but strongly correlated core node, with short efficient paths and clustering coefficients significantly higher than random baseline models. Compared to similar ER, BA, and WS graphs, this network is not completely random and cannot be fully explained by a single classical model. Instead, it combines inequalities between nodes with high local correlation. Therefore, the analysis of open-ended questions supports the conclusion that influence in the voting network is shaped by both reputation and mediation.

References

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